

Matt Schwarz

mattschwarz.xyz | github.com/mjschwarz

EDUCATION

Vanderbilt University

Majors in Computer Science and Medicine, Health, & Society; Minor in Economics

May 2024

Nashville, TN

- GPA: 3.92 / 4.00 (Cum Laude)
- SAT: 1570 / 1600 (Math: 800 / 800)

EXPERIENCE

Walmart Global Technology

Software Engineer II

Bentonville, AR

July 2024 - Present

- Deployed a pipeline to identify items needing substitutes, avoiding over \$600,000 in lost sales and saving over 2,000 weekly merchant workhours in the first 3 months after launch
- Developed key features for an industry-leading AI evaluation framework, increasing test coverage of agentic capabilities to over 80% and decreasing developer testing time from 1-2 hours per deployment to several minutes
- Spearheaded an initiative to add hundreds of business-performance metrics to Wally, an AI agent focused on streamlining merchant's day-to-day tasks, reducing merchants' time spend per metric from 3 minutes to 30 seconds
- Refined Wally's prompt to optimize token utilization and scalability, decreasing per-question LLM costs by thousands of tokens and improving the AI agent's accuracy scores by over 10% even as it scaled to dozens of tools
- Enhanced the embeddings process for Wally's merchant knowledge base by extracting additional metadata from source materials, improving the AI agent's accuracy scores by 15%

Software Engineer Intern

May 2023 - August 2023

- Identified a business need for more rigorous e-commerce analytics through discussions with primary stakeholders
- Built a dashboard using Python and PostgreSQL, improving merchants' ability to analyze site performance
- Enhanced algorithms for three key performance indicators, increasing insight into actionable business opportunities

Ricoh USA

Software Engineer Intern

Remote

June 2022 - January 2023

- Developed programs to populate customer bundles with the latest versions of software, reducing manual error
- Streamlined developers' project setup workflow using Groovy scripting to automate repetitive version control tasks
- Created automated testing builds using Jenkins, decreasing manual testing time per release by weeks

Active Galactic Nuclei

Research Assistant

Vanderbilt University

August 2020 - May 2021

- Collaborated with four team members in investigating supermassive black hole phenomena
- Analyzed spectral data sets of quasar observations by writing Python programs to perform curve fitting
- Generated spectral decomposition plots and other reports using specialized scientific packages

PROJECTS

Galactic Showdown (Laser Tag VR) | C#, Unity, Photon

- Implemented real-time networking for up to four players as well as destructible objects using Photon
- Wrote custom scripts in C# to handle networking of hit registration
- Designed a head- and hand-tracked VR avatar and sci-fi themed arena environment in Unity

LEADERSHIP AND EXTRACURRICULARS

Code Ignite

Teaching Assistant

Vanderbilt University

September 2021 - May 2024

- Taught ~10 middle school students computer science and problem-solving principles on a weekly basis
- Developed a Java curriculum for middle and high school students alongside club leadership

SKILLS AND INTERESTS

Languages: Python, Java, SQL, C/C++, JavaScript

Tools and Frameworks: Inspect AI, Kubernetes, Git, PostgreSQL, Docker, React, Jenkins

Interests: Triathlon, SCUBA diving, Weightlifting, Hiking (Eagle Scout), Beta Theta Pi